CT-1 (CN)

1. Differentiate between OSI and TCP/IP reference Models in at least six aspects with neat illustrations of the models

Answer:

OSI(Open System Interconnection)	TCP/IP(Transmission Control Protocol / Internet Protocol)
1. OSI is a generic, protocol independent standard, acting as a communication gateway between the network and end user.	1. TCP/IP model is based on standard protocols around which the Internet has developed. It is a communication protocol, which allows connection of hosts over a network.
2. In OSI model the transport layer guarantees the delivery of packets.	2. In TCP/IP model the transport layer does not guarantees delivery of packets. Still the TCP/IP model is more reliable.
3. Follows vertical approach.	3. Some times it follows horizontal approach.
4. OSI model has a separate Presentation layer and Session layer.	4. TCP/IP does not have a separate Presentation layer or Session layer.
5. OSI is a reference model around which the networks are built. Generally it is used as a guidance tool.	5. TCP/IP model is, in a way implementation of the OSI model.
6. Network layer of OSI model provides both connection oriented and connectionless service.	6. The Network layer in TCP/IP model provides connectionless service.
7. OSI model has a problem of fitting the protocols into the model.	7. TCP/IP model did not fit any other protocol stacks.
8. Protocols are hidden in OSI model and are easily replaced as the technology changes.	8. In TCP/IP replacing protocol is not easy.
9. OSI model defines services, interfaces and protocols very clearly and makes clear distinction between them. It is protocol independent.	9. In TCP/IP, services, interfaces and protocols are not clearly separated. It is also protocol dependent.
10. It has 7 layers	10. It has 4 layers

2. In a circuit-switched network, the call setup time is 3 seconds and data transmission time is 9 seconds. Find the total delay involved.

Answer:

According to the text, "Total delay includes: Time to establish the connection (setup time) [and] Time to send the data (transmission time)".

- **Given:**
  - Call setup time = 3 seconds
  - Data transmission time = 9 seconds
- **Calculation:**
  - Total Delay = Call setup time + Data transmission time
  - Total Delay = 3s + 9s = **12 seconds**

3. Compare twisted pair, coaxial, and optical fiber cables with respect to bandwidth, attenuation, and cost with respect to their applications and limitations

Answer:

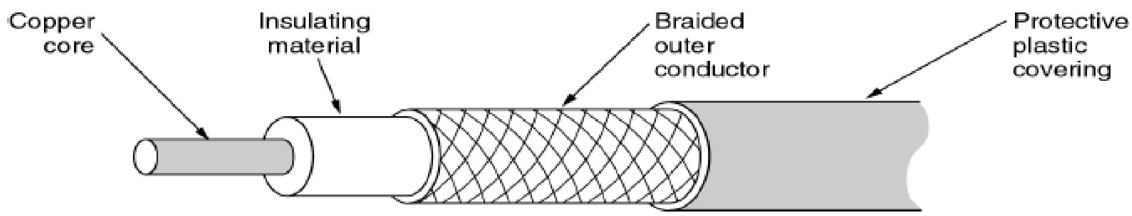
- **Twisted Pair:**
  - **Bandwidth:** "Bandwidth is low when compared with Coaxial Cable."
  - **Cost:** "It is lightweight, cheap, can be installed easily."
  - **Attenuation:** "Can only be used for shorter distances because of attenuation."
  - **Limitations:** "Provides less protection from interference."
- **Coaxial Cable:**
  - **Bandwidth:** "Bandwidth is high... Modern cables have a bandwidth of close to 1 GHz."
  - **Cost:** "Difficult to install and expensive when compared with twisted pair."
  - **Attenuation/Distortion:** "Data transmission without distortion."
  - **Applications:** "Widely used for Cable TV, WAN (Internet over cable)."
- **Optical Fiber:**
  - **Bandwidth:** "Supports high bandwidth." and "Can transmit data at very high speed."
  - **Cost:** "Fragile and expensive."
  - **Attenuation:** "There is less signal attenuation."
  - **Applications:** "Used for longer distances... Currently available single-mode fibers can transmit data at 50 Gbps for 100 km."

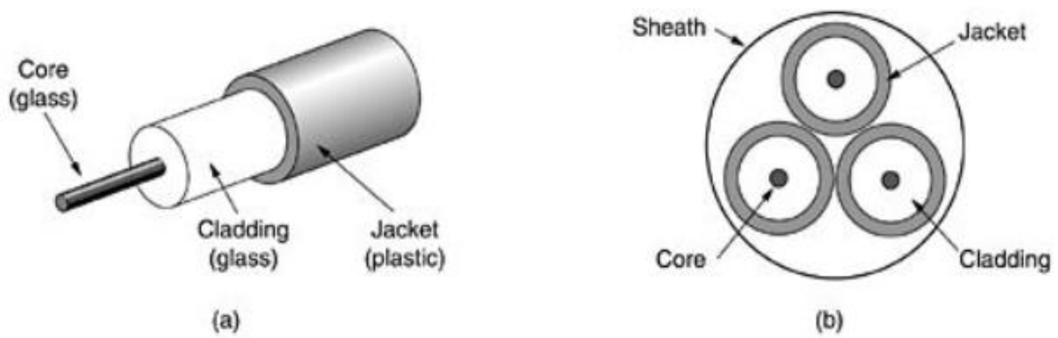
or

Comparison of Transmission Media

Feature	Twisted Pair Cable	Coaxial Cable	Optical Fiber Cable
Bandwidth	**Low** when compared with Coaxial Cable. It supports several megabits/sec.	**High.** Modern cables have a bandwidth of close to **1 GHz.**	**Very High.** It can transmit data at very high speeds (e.g., 50 Gbps).
Attenuation	**High.** It can only be used for shorter distances because of attenuation.	**Moderate.** It allows data transmission without distortion and spans longer distances than twisted pair due to better shielding.	**Low.** There is less signal attenuation, allowing transmission over very long distances (e.g., 100 km without amplification).
Cost	**Low.** It is lightweight, cheap, and easy to install.	**Moderate/High.** It is expensive when compared with twisted pair.	**High.** The cable is expensive.

Applications	Widely used for **telephone communications** and **Ethernet networks** (LANs).	Widely used for **Cable TV** and **WANs** (Internet over cable).	Used for **long-distance** communication and single-mode fibers are used for high-speed transmission.
Limitations	Provides **less protection from interference** and is prone to noise/crosstalk.	**Difficult to install** and a single cable failure can fail the entire network.	**Fragile**, and installing/maintaining it is difficult.





4. Identify the type of topology on the basis of the following:

(i) Since every node is directly connected to the server, a large amount of cable is needed which increases the installation cost of the network.

Answer: Star Topology

- **Justification:** The text describes Star Topology where "All devices are connected to a centralized networking device" (server/switch). It lists the disadvantages as: "Cost of installation is high... Too many cables make the network messy."

(ii) It has a single common data path connecting all the nodes

Answer: Bus Topology

- **Justification:** The text states, "Bus topology is a network type in where every computer and network device is connected to single cable... This central cable, called bus or backbone."
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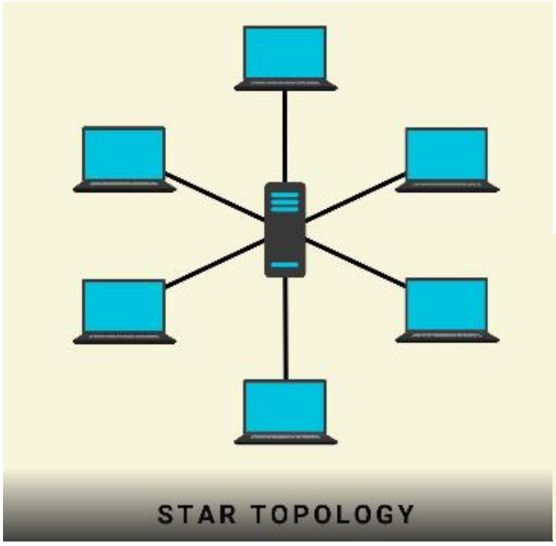
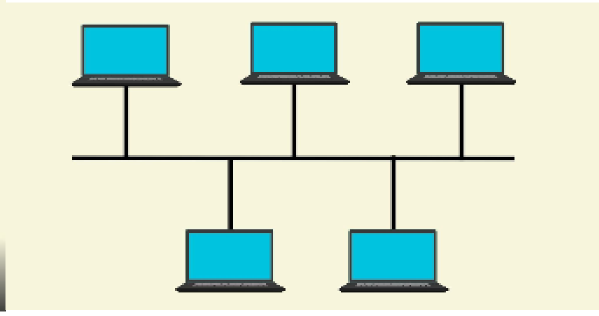




5. Illustrate the functioning of protocols present in the Transport and Internet Layers of TCP/IP Reference model and justify their usage for respective applications.

Answer:

Internet Layer Protocols:

- **IP (Internet Protocol):** "It is the primary protocol in the internet layer... It is responsible for the transmission of data packets called datagram's from the source to the destination host."
- **ARP (Address Resolution Protocol):** "Its main responsibility is to find the physical address of the host using the internet address or IP address."
- **ICMP (Internet Control Message Protocol):** "Is a mechanism used by hosts and gateways to send notification of datagram problems back to the sender."
- **IGMP (Internet Group Message Protocol):** "It is used for the transmission of data to a group of networks simultaneously. For eg. online streaming."

Transport Layer Protocols:

- **TCP (Transmission Control Protocol):** "It is a reliable connection-oriented protocol that transmits data from the source to the destination machine without any error." It provides "Data sequencing" and "Error Checking and Recovering."
    - **Usage:** "Internet searching(www), Email delivery, File transfer etc."
- **UDP (User Datagram Protocol):** "It is a message-oriented protocol that provides a simple unreliable, connectionless, unacknowledged service." It is "Faster then TCP because it does not provide any feedback."
    - **Usage:** "Online streaming movies, songs,games, voice over notes etc."

6. A message of size 1 MB is transmitted over a link with a bandwidth of 2 Mbps. Calculate the transmission time.

Answer:

- **Given:**
  - Message Size = 1 MB (Megabyte)
  - Bandwidth = 2 Mbps (Megabits per second)

- **Conversion:**
  - $1 \text{ Byte} = 8 \text{ bits}$
  - $1 \text{ MB} = 1{,}000{,}000 \text{ Bytes} \approx 8{,}000{,}000 \text{ bits} = 8 \text{ Megabits (Mb)}$
- **Calculation:**

$$\text{Transmission Time} = \frac{\text{Message Size}}{\text{Bandwidth}}$$

$$\text{Transmission Time} = \frac{8 \text{ Mb}}{2 \text{ Mbps}} = \mathbf{4 \text{ seconds}}$$

7. Compare circuit switching and packet switching with respect to bandwidth utilization, delay, and reliability. In a circuit-switched network, a dedicated link of 1 Mbps is reserved for 10 seconds, but actual data transmission occurs only for 4 seconds. Calculate the bandwidth utilization percentage.

Answer:

Comparison:

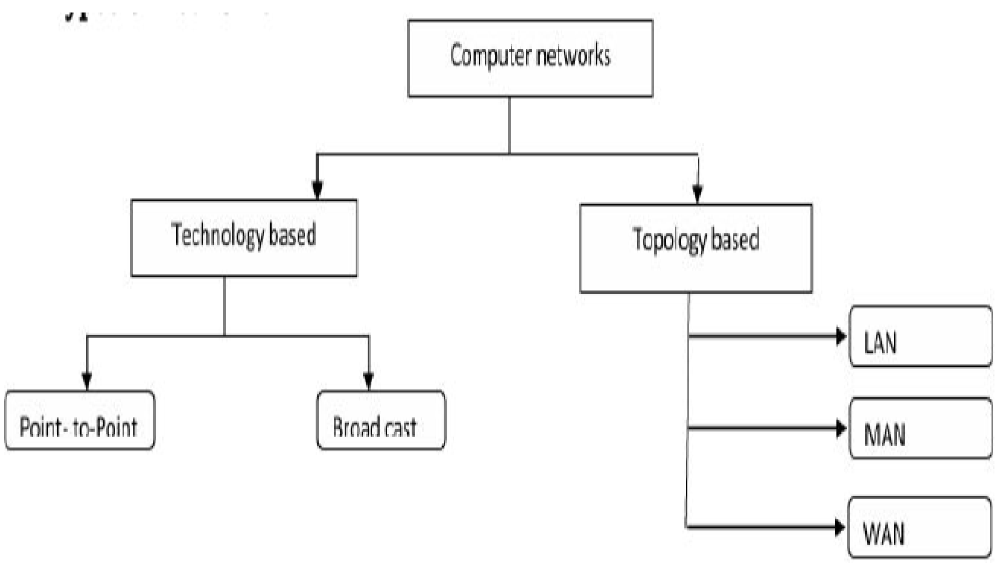
- **Bandwidth Utilization:**
  - **Circuit Switching:** "Resources are allocated during the entire duration of the connection. These resources are unavailable to other connections," making it less efficient.
  - **Packet Switching (Datagram):** "Resources are allocated only when there are packets to be transferred," making efficiency better.
- **Delay:**
  - **Circuit Switching:** "Have very low delay during data transfer... There is no processing delay at switches because the path is already reserved."
  - **Packet Switching:** "There may be greater delay... each packet may experience a wait at a switch before it is forwarded."

- **Reliability:**
  - **Circuit Switching:** Provides a dedicated path and is generally connection-oriented.
  - **Packet Switching:** Can be connectionless (Datagram) where the switch "does not keep information about the connection state" and there is "No guaranty to received all data" (in UDP).

Calculation:

- **Given:**
  - Reserved Duration = 10 seconds
  - Actual Transmission Duration = 4 seconds
- **Bandwidth Utilization Percentage:**
  - Utilization = $\left(\frac{\text{Time Used}}{\text{Total Reserved Time}}\right) \times 100$
  - Utilization = $\left(\frac{4}{10}\right) \times 100$ = **40%**

8. List the different ways of categorizing networks giving two examples of each category



According to the "Types of Networks" section in the text:

1. **Technology based:**
   - **Examples:** "Point-to-Point", "Broadcast"
2. **Topology based (or Geographical area):**

- **Examples:** "LAN (Local Area Network)", "WAN (Wide Area Network)" (also MAN, PAN)

3. **Scale based (Physical Size):**
   - **Examples:** "Personal area network" (1m), "Metropolitan area network" (10km)